ABHISHEK SHRIVASTAVA

Senior Data Engineer | Data Platforms · Snowflake · dbt · Spark | Production AI/ML

Event-Driven & Metadata-Driven Pipelines · Dimensional Modeling · Governance & Observability · RAG / LLM Systems

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7 years building high-scale, production data platforms — event-driven and metadata-driven pipelines, lakehouse and warehouse architectures, dimensional modeling, and the data-quality, governance, and observability that keep them trustworthy. Expert in **SQL, Snowflake, dbt, Spark, and AWS**, with a track record of cutting pipeline latency from days to minutes and scaling to billions of events/day. Now extends that same platform rigor into production AI/ML — RAG systems built with evaluation, grounding, and observability.

SKILLS

Data Engineering: Event-driven & metadata-driven architecture, ETL/ELT, dimensional modeling (star schema, SCDs), lakehouse (Iceberg), data warehousing, streaming, orchestration, data quality, governance, lineage & auditability

Databases & SQL: Expert SQL (window functions, CTEs, query optimization), Snowflake (performance tuning, modeling), Redshift, PostgreSQL, Hive

Cloud & Tooling: AWS (Glue, S3, Lambda, EventBridge, Redshift, EMR, DMS), dbt, Airflow, Docker/Kubernetes, GitHub Actions, CI/CD, FastAPI

Big Data & Languages: Spark, PySpark, Snowpark, Iceberg, Kafka, distributed systems · Python, SQL

AI/ML: Production RAG, hybrid retrieval, cross-encoder reranking, embeddings (sentence-transformers), RAG evaluation (recall@k, MRR), Model Context Protocol (MCP), LLM integration, pgvector, PyTorch

PROFESSIONAL EXPERIENCE

Senior Data Engineer — EDP Platform, MUFG Bengaluru Jan 2024 – Present

AWS · Spark · Python · SQL · Snowflake concepts · Glue · EventBridge · Iceberg · Redshift · Lambda

- Designed serverless, **event-driven data pipelines** (AWS DMS → S3 → EventBridge → Glue → Iceberg → Redshift) powering near real-time model-risk reporting across multiple risk domains.
- Built **metadata-driven orchestration** (Postgres job control governing stage → transform → enterprise → info-mart stages) that generates pipelines from config — scaling to 43 workflows (~170 jobs) with a ~3-minute, config-driven deploy via Lambda + boto3 CI/CD.
- **Cut pipeline latency from hours to minutes** by removing manual hand-offs and standardizing validations — row-hash reconciliation, schema-evolution guards(glue catalog), and versioned-partition reproducibility for audit.
- Migrated historical datasets from Redshift to an **Iceberg lakehouse** for time-travel, auditability, and cost-efficient retrieval; hardened lineage and versioning for reproducible regulatory reporting.
- Built an internal **Model Context Protocol (MCP) server** exposing read-only Hive operations as governed tools for AI assistants — enabling prompt-based SQL workflows against the warehouse with read-only enforcement and data-egress controls.

Data Engineer — Data Intelligence, SailPoint Technologies (US Team / Pune) Mar 2022 – Jan 2024

Snowflake · dbt · Airflow · Python · Spark / Snowpark · AWS · CI/CD

- Designed **Snowflake data models** powering IdentityAI, processing **billions of events/day**; built dimensional models and governance patterns for analytics at scale.
- Built a **"Pipeline Factory"** to bootstrap new Airflow DAGs from config files — reduced onboarding time ~70% and standardized data-quality and governance patterns across teams.
- Drove **cost optimization** by replacing Flink with Spark-based Jaccard computation; multi-tenant design over billions of rows/day, saving in the order of millions annually.
- Led a **Snowpark + dbt** POC centralizing testing and data-quality checks; built CI/CD and migrated internal packages to AWS CodeArtifact for versioned distribution.
- Owned **on-call** — added alerting/observability, triaged incidents, and coordinated infra fixes for high SLA adherence.

Deputy Manager — Data Science & Engineering, DCB Bank Mumbai Oct 2019 – Mar 2022

Hadoop (Cloudera/Hortonworks) · Hive · Pentaho · Python · SQL

- Engineered end-to-end pipelines for Internet/Mobile Banking; migrated workloads to **Hadoop/Hive** for scalable reporting and lower TCO.
- Automated **RBI compliance & operational reporting** (financial inclusion, NEFT/RTGS, Video-KYC, ATM dispensations, gold-loan exposure).
- Delivered daily MIS & executive dashboards with validation and freshness/quality alerting; **mentored 6+ engineers** across the Big Data stack.

SELECTED PROJECTS

AI Incident Copilot — Production RAG for incident root-cause analysis

- Built a production RAG pipeline: **hybrid retrieval** (SQL time-filter + pgvector dense search), cross-encoder reranking, and a confidence quality-gate that suppresses low-grounding answers (no context → no LLM call → no hallucination). FastAPI, Dockerized, CI/CD, 267 tests.
- Built a 31-question **evaluation harness** (recall@k, MRR) — measured perfect lookup retrieval (recall@5 = 1.0) and surfaced a multi-hop gap (recall@5 = 0.33), diagnosed to single-pass retrieval with scoped fixes (iterative retrieval, chain-aware reranking).
- Chose a **deterministic pipeline over an agent** for the core flow — incident triage has known steps, and trust under pressure beats autonomous cleverness.

WRITING

From Days to Minutes: Automating Data Pipelines with AWS Event-Driven Architecture — Medium

EDUCATION & CERTIFICATIONS

- **AWS Certified Cloud Practitioner** (2023)
- **Springboard** — Data Science (Program - 2021)
- **PG Diploma, Big Data Analytics** — CDAC (2019)
- **B.E. Computer Science** — RGPV (2018)